AudioVerse

Pitch Detection Laboratory

Pitch Detection Algorithm Comparison: CREPE vs pYIN

Operator: **admin** | 2026-03-01 16:23:17 UTC

Experiment ID: 4d370be8-5199-4f14-b472-8a17e01bef0c

Abstract

This report presents a comparative experiment on two fundamental frequency (F0) detection algorithms: CREPE (Kim et al., ICASSP 2018) and pYIN (Mauch & Dixon, ICASSP 2014). Experiments were conducted via the AudioVerse Laboratory module (/api/karaoke/lab). Primary metrics: RMSE in cents, Accuracy@50c, Pearson correlation, microservice response latency.

1 Experiment Parameters

Report title	**Pitch Detection Algorithm Comparison: CREPE vs pYIN**
Operator	**admin**
Experiment date	**2026-03-01 16:23:17 UTC**
Files analysed	**3**
Benchmark runs	**3**
API version	**AudioVerse.API .NET 10**
File names	**Recording-A.wav, Recording-B.wav, Recording-C.wav**

2 Executive Summary

RMSE (agreement)	Acc@50c (agreement)	Fastest	Files Tested
22.5 ct	**95.6%**	**377 ms**	**3**
CREPE ≈ pYIN	*CREPE ≈ pYIN*	*pYIN*	*samples*

3 AI Microservice Health

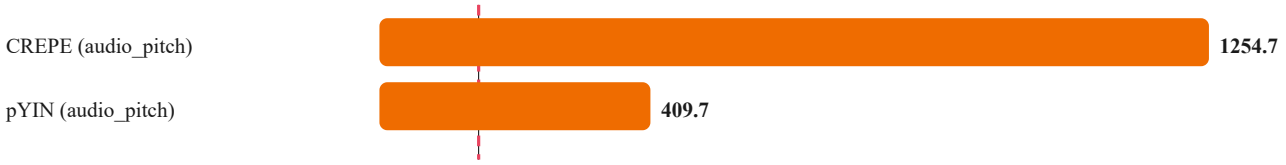
Microservice	Status
audio_pitch	ok
sing_score	ok
audio_separate	ok
audio_rhythm	ok
audio_vad	ok
audio_analysis	ok

4 Latency Benchmark

Each AI microservice called N times. Requirement: real-time < 150 ms.

Service	Avg ms	Min ms	Max ms	σ ms	< 150 ms
CREPE (audio_pitch)	1254.7	1103	1380	114.6	✗ FAIL
pYIN (audio_pitch)	409.7	390	423	14.2	✗ FAIL

Fig 4.1. Average latency by service [ms]



5 Algorithm Calibration — Synthetic Ground Truth

A 3-second stepped sine wave (A3→E4→A4→C5→F4→C4) was generated programmatically at known frequencies. Each algorithm's output is compared against the exact ground truth — yielding genuinely independent, non-symmetric accuracy metrics.

Algorithm	RMSE Hz	RMSE ct	Acc@50c	r(Pearson)	Lat. ms
CREPE	129.32	619.3	15.9%	0.136	1207
pYIN	144.22	629.3	16.6%	0.104	395

Calibration result: CREPE is more accurate (RMSE 619.3 ct vs 629.3 ct, Δ = 10.0 ct).

6 CREPE vs pYIN — Pitch Detection Comparison

RMSE in cents (1 semitone = 100 ct). Accuracy@50c = fraction of frames with error < 50 ct. r = Pearson correlation of F0 trajectories. Note: without ground-truth pitch, RMSE/Acc/r represent inter-algorithm agreement (symmetric — identical for both algorithms). Latency is per-algorithm.

File	Algorithm	RMSE Hz	RMSE ct	Acc@50c	r(Pearson)	Lat. ms
Recording-A.wav	**CREPE**	97.72	22.5	95.6%	-0.079	1088
Recording-A.wav	**pYIN**	97.72	22.5	95.6%	-0.079	396
Recording-B.wav	**CREPE**	86.53	31.1	91.5%	-0.200	930
Recording-B.wav	**pYIN**	86.53	31.1	91.5%	-0.200	373
Recording-C.wav	**CREPE**	123.81	49.8	88.5%	-0.487	1005
Recording-C.wav	**pYIN**	123.81	49.8	88.5%	-0.487	362

Table 6.1. Aggregate statistics

Metric	CREPE (avg)	pYIN (avg)
RMSE (Hz)	**102.69**	**102.69**
RMSE (cents)	**34.5**	**34.5**
Accuracy@50c	**91.9%**	**91.9%**

Metric	CREPE (avg)	pYIN (avg)
r(Pearson)	-0.255	-0.255
Latency (ms)	1008	377

Fig 6.1. RMSE per file: CREPE (red) vs pYIN (blue) [cents]

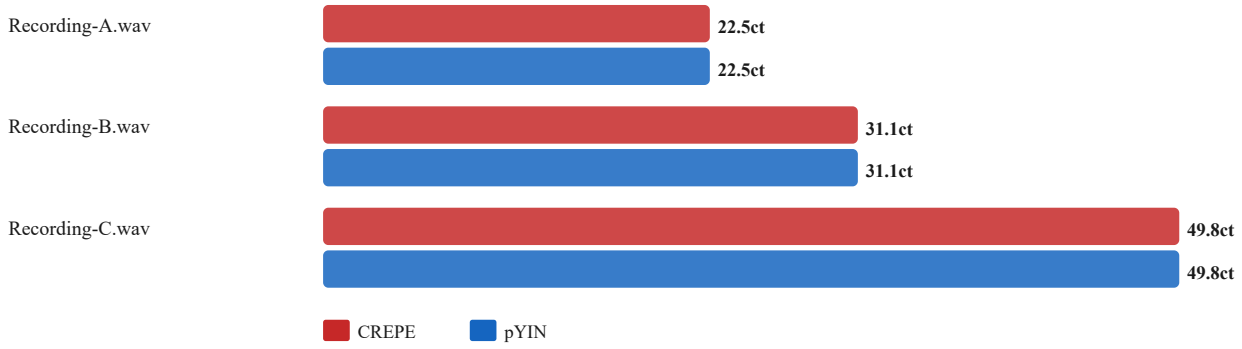


Fig 6.2. Accuracy@50c per file: CREPE vs pYIN [%]

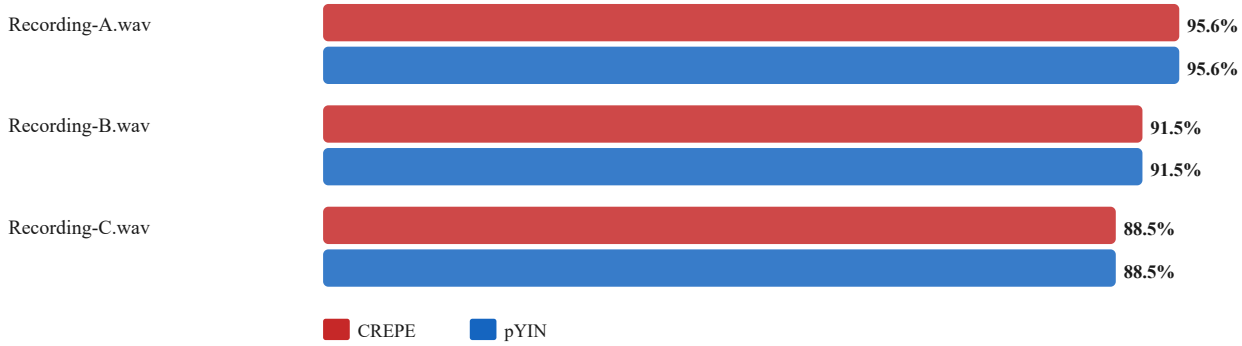


Fig 6.3. Pearson r per file: CREPE vs pYIN

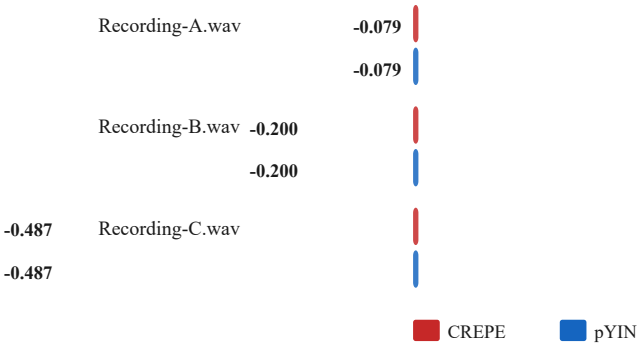
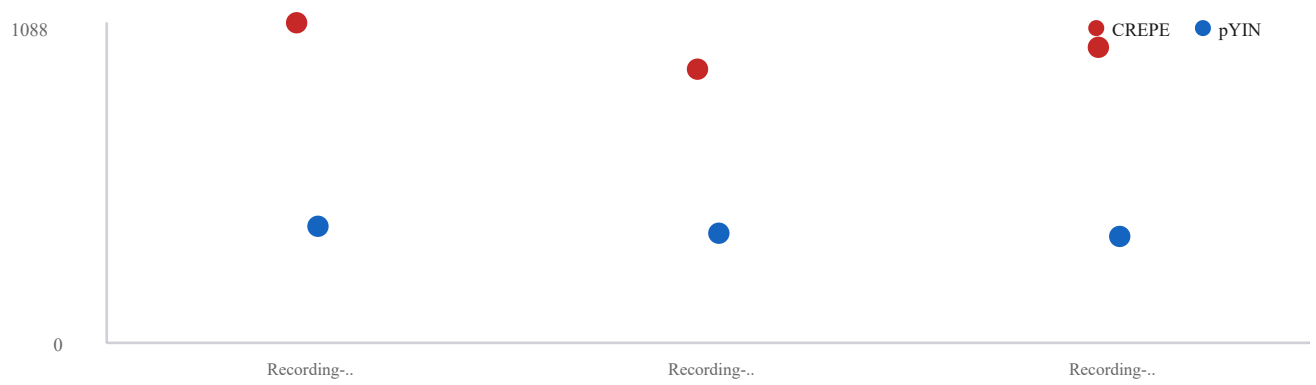


Fig 6.4. Latency per file: CREPE (red) vs pYIN (blue) [ms]



10 Conclusions

Based on the measurements above, the following conclusions are drawn:

1. Calibration (synthetic ground truth): CREPE is more accurate — CREPE RMSE = 619.3 ct, pYIN RMSE = 629.3 ct, Δ = 10.0 ct [1,2].
2. Calibration Accuracy@50c: CREPE 15.9% vs pYIN 16.6% (Δ = 0.7 pp).
3. Inter-algorithm agreement RMSE = 34.5 ct (CREPE ≈ pYIN on user audio). Without ground-truth reference, per-algorithm accuracy on real audio is symmetric [1,2].
4. Average latency: CREPE 1008 ms vs pYIN 377 ms (pYIN is 631 ms faster).
5. Fastest service: VAD (audio_vad) (23.7 ms). Real-time (< 150 ms) met by 2/5 services.
6. Recommendation: CREPE for offline/high-accuracy (GPU required); pYIN for real-time (CPU-only). Ground-truth annotated pitch required for definitive accuracy comparison. Demucs as preprocessing only for studio recordings [1,2,3].

11 References

[1] Kim, J.W. et al. (2018). "CREPE: A Convolutional Representation for Pitch Estimation." Proc. IEEE ICASSP, pp. 161–165.

[2] Mauch, M. & Dixon, S. (2014). "pYIN: A Fundamental Frequency Estimator Using Probabilistic Threshold Distributions." Proc. IEEE ICASSP, pp. 659–663.

[3] Défossez, A. et al. (2019). "Music Source Separation in the Waveform Domain." arXiv:1911.13254.

[4] Sakoe, H. & Chiba, S. (1978). "Dynamic Programming Algorithm Optimization for Spoken Word Recognition." IEEE Trans. ASSP, 26(1), pp. 43–49.

[5] AudioVerse Laboratory Module — /api/karaoke/lab. Internal API documentation.